Design History File (DHF)

Generated automatically. Contents listed below.

Core Documents

- Software Requirements Specification
- Risk Management Log
- Verification Report
- Traceability Matrix

Notes

All documents are generated from controlled GitHub Issues and records. Manual edits are not permitted.

Software Requirements Specification (SRS)

Generated automatically. Do not edit manually.

SRS-100

Requirement text

12-lead

IEC 62304 Safety Class

A

Acceptance criteria

1: filter 2: module 3: module throws an error

SRS-002

Requirement text

The system shall log user authentication failures

IEC 62304 Safety Class

B

Acceptance criteria

deterministic pass/fail

SRS-002

Requirement text

The system needs to pass criteria.

IEC 62304 Safety Class

B

Acceptance criteria

pass/fail

SRS-001

Requirement text

The system shall reject any user input that does not conform to the defined input format.

IEC 62304 Safety Class

B

Acceptance criteria

- Invalid input is rejected with an error message.
- No processing occurs on invalid input.
- Valid input continues to be processed normally.

Risk Management Log (ISO 14971)

Generated automatically. Do not edit manually.

R-002

Hazard / Hazardous Situation

12-lead does not fishing within 30 seconds

Potential Harm

The system crashes

Risk Controls

Not yet defined

Risk Status

unspecified

R-100

Hazard / Hazardous Situation

The leads become too hot

Potential Harm

The patient gets burnt

Risk Controls

Not yet defined

Risk Status

unspecified

R-001

Hazard / Hazardous Situation

Invalid user input could be processed without validation, leading to incorrect system behavior.

Potential Harm

Incorrect output or system malfunction affecting the user.

Risk Controls

Not yet defined

Risk Status

unspecified

R-001

Hazard / Hazardous Situation

Invalid user input may be processed by the system.

Potential Harm

Incorrect system behavior or output.

Risk Controls

Not yet defined

Risk Status

unspecified

Verification Report

Generated automatically. Do not edit manually.

T-002

Traces to Requirement(s)

SRS-002

Test Procedure

Provide invalid input to the system. Observe system response. Provide valid input.

Expected Result

The 12-lead completes within 30 seconds

T-100

Traces to Requirement(s)

SRS-100

Test Procedure

Measure the time to complete the procedure and the temperature it reaches within that time.

Expected Result

The procedure takes less than 30 seconds and the temperature does not increase above 15 degrees throughout the process

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T-001

Traces to Requirement(s)

SRS-002

Test Procedure

1. Provide invalid input to the system.
2. Observe system response.
3. Verify that invalid input is rejected.

Expected Result

The system rejects invalid input and does not proceed with processing.

T-001

Traces to Requirement(s)

SRS-001

Test Procedure

1. Provide invalid input to the system.
2. Observe system response.
3. Provide valid input.

Expected Result

- Invalid input is rejected.
- Valid input is accepted and processed.

Requirement ID,Risk IDs,Test IDs “SRS-100”,“ “,”T-100” “SRS-002”,“ “,”T-002” “SRS-001”,“R-001”,“T-001”